

Ch1: Searching Basics

Chapter Goal

Understand what searching is, what a **target** is, the basic searching idea, and the **main types of searching** used in C.

Slide 1 — What is Searching?

Searching

Searching is the process of finding a particular value from a collection of values.

Example:

Numbers: 12 25 8 40 17



Target = 8

The computer checks the available values to determine whether the **target** is present.

Real-Life Example

 You have many books and want to find "**C Programming**".

You search through the books until you find the one you need.

Searching = Finding a required item from a collection.

Slide 2 — Target, Found & Not Found

When searching, we usually have **two important things**:

Target

The value we are looking for.

Example:

10 25 40 15 30
 ↑
 Target = 40

Search Result

If the target exists:

Found

If it doesn't exist:

Not Found

Example

Search for 15:

10 → 25 → 40 → 15 → 30
 
 FOUND 

Search for 50:

10 → 25 → 40 → 15 → 30

50 
NOT FOUND

Slide 3 — Types of Searching

There are different ways to search through data.

Two important searching methods

1 Linear Search

Checks values **one by one**.

10 → 25 → 40 → 15 → 30

↓ ↓ ↓

Check Check Check...

 Works even when the data is **not sorted**.

2 Binary Search

Repeatedly **divides the search area into two parts**.

10 20 30 40 50 60 70

↓

Middle

 Requires the data to be **sorted**.

 **Quick Memory**

Linear → One by One

Binary → Divide into Two

Slide 4 — Linear vs Binary Search

	Linear Search	Binary Search
How it searches	One by one	Divides the data
Data required	Can be unsorted	Must be sorted
Basic idea	Check each value	Check the middle and narrow down

Beginner
example

Searching a name in an
unsorted list

Finding a word in a dictionary

Think About It

Suppose you have:

2 5 8 12 16 20 25

 Find 20.

Linear Search:

2 → 5 → 8 → 12 → 16 → 20 

Binary Search:

Start around the middle → decide which half to search → continue.

 **Different situations can need different searching methods.**

Slide 5 — Searching Basics Recap

 **Remember These 5 Things**

 **Searching** → Finding a value in a collection.

 **Target** → The value we want to find.

 **Found** → Target exists.

 **Not Found** → Target doesn't exist.

 **Types we'll learn:**

- Linear Search → **One by One**
- Binary Search → **Divide & Search**

 **Memory Trick**

LINEAR = Look one by one 👁️

BINARY = Break into two ✂️

🏆 **LEVEL CLEARED!**

You now know **what searching is and the two main searching methods.**

QUESTIONS:-

🔍 Ch1 — Searching Basics Quiz

Q1. 🧠 MCQ

What is the main purpose of searching?

- A) To arrange values
- B) To find a required value
- C) To delete values
- D) To add values

💡 **Hint:** Think about what you do when you are looking for something specific.

📖 **Glossary:**

- 🔍 **Searching** — Finding something you are looking for.

✅ **Answer: B) To find a required value**

Q2. 🎯 MCQ

In searching, what do we call the value that we are looking for?

- A) Index
- B) Target
- C) Array
- D) Loop

 **Hint:** The name given to the value you want to find is related to aiming at something.

 **Glossary:**

-  **Target** — The value we are looking for.

 **Answer: B) Target**

Q3. MCQ

Which of the following is a type of searching?

- A) Linear Search
- B) Bubble Search
- C) Selection Search
- D) Insertion Search

 **Hint:** One option is a searching method that checks values one by one.

 **Glossary:**

-  **Linear Search** — A searching method that checks values one by one.

 **Answer: A) Linear Search**

Q4. Fill in the Blank

A value that we want to find in a collection is called the _____.

 **Hint:** Think about the value you are aiming to find.

 **Glossary:**

-  **Target** — The value we want to find.

 **Answer: Target**

Q5. Code Fill-in-the-Blank

Complete the missing code to check whether the target value 25 is present.

```
#include <stdio.h>
```

```
int main() {  
    int numbers[] = {10, 15, 25, 30};  
    int target = 25;  
  
    if(numbers[2] == _____)  
        printf("Found");  
  
    return 0;  
}
```

 **Hint:** The missing word should represent the value we are searching for.

Glossary:

-  **target** — The value being searched for.

 **Answer:** target